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Safety data sheet

BASF safety data sheet. This is a translation of the country-specific safety data sheet into a language other than that required by law. This document does not replace the safety data sheet provided according to Regulation (EC) No 1907/2006.

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Product: **Basonat® HI 100 NG CN**

(ID no. 30675011/SDS_GEN_DE/EN)

Date of print 07.07.2025

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

Basonat® HI 100 NG CN

Chemical name: Hexane, 1,6-diisocyanato-, oligomers
CAS Number: 28182-81-2

REACH registration number: 01-2119485796-17-0004, 01-2119485796-17-0015

1.2. Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: Intermediate (isolated), Hardener for coating materials or adhesives for industrial or professional use, Isocyanate used in the manufacturing of polyurethanes
Uses advised against: All consumer uses are strongly advised against., The hazardous properties of the substance require safety measures which can, in principle, not be sufficiently ensured in the home worker sector.

For the detailed identified uses of the product see appendix of the safety data sheet.

1.3. Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY
Global Business Unit Resins and Additives

Telephone: +49 621 60-0
E-mail address: ed-psr@basf.com

1.4. Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

SECTION 2: Hazards Identification

2.1. Classification of the substance or mixture

According to Regulation (EC) No 1272/2008 [CLP]

Acute Tox. 4 (Inhalation - mist)	H332 Harmful if inhaled.
Skin Sens. 1	H317 May cause an allergic skin reaction.
STOT SE 3	H335 May cause respiratory irritation.

For the classifications not written out in full in this section the full text can be found in section 16.

2.2. Label elements

According to Regulation (EC) No 1272/2008 [CLP]

Pictogram:

Signal Word:

Warning

Hazard Statement:

H332	Harmful if inhaled.
H317	May cause an allergic skin reaction.
H335	May cause respiratory irritation.

Precautionary Statements (Prevention):

P280	Wear protective gloves.
P271	Use only outdoors or in a well-ventilated area.
P260	Do not breathe mist or vapour.
P272	Contaminated work clothing should not be allowed out of the workplace.

Precautionary Statements (Response):

P312	Call a POISON CENTER or physician if you feel unwell.
P304 + P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P303 + P352	IF ON SKIN (or hair): Wash with plenty of soap and water.
P333 + P311	If skin irritation or rash occurs: Call a POISON CENTER or physician.
P362 + P364	Take off contaminated clothing and wash it before reuse.

Precautionary Statements (Storage):

P403 + P233	Store in a well-ventilated place. Keep container tightly closed.
P405	Store locked up.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.

Labeling of special preparations (GHS):

EUH204: Contains isocyanates. May produce an allergic reaction. EUH208: May produce an allergic reaction. Contains: hexamethylene-di-isocyanate
As from 24 August 2023 adequate training is required before industrial or professional use.

Hazard determining component(s) for labelling: HDI oligomers, isocyanurate

2.3. Other hazards

According to Regulation (EC) No 1272/2008 [CLP]

No specific dangers known, if the regulations/notes for storage and handling are considered. Product does not contain a substance above legal limits included in the list established in accordance with Article 59(1) of Regulation (EC) No 1907/2006 for having endocrine disrupting properties or is identified to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605. The product does not contain a substance above legal limits fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

SECTION 3: Composition/Information on Ingredients

3.1. Substances

Chemical nature

polyfunctional polyisocyanate

Regulatory relevant ingredients

HDI oligomers, isocyanurate

Content (W/W): >= 95 % - <= 100 %	Acute Tox. 4 (Inhalation - mist)
CAS Number: 28182-81-2	Skin Sens. 1
	STOT SE 3 (irr. to respiratory syst.)
	H332, H317, H335

hexamethylene-di-isocyanate

Content (W/W): >= 0 % - <= 0,3 % CAS Number: 822-06-0 EC-Number: 212-485-8 INDEX-Number: 615-011-00-1	Acute Tox. 4 (oral) Acute Tox. 1 (Inhalation - mist) Skin Corr./Irrit. 2 Eye Dam./Irrit. 2 Resp. Sens. 1 Skin Sens. 1 STOT SE 3 (irr. to respiratory syst.) H319, H315, H330, H302, H334, H317, H335 Specific concentration limit: Skin Sens. 1: >= 0,5 % Resp. Sens. 1: >= 0,5 %

For the classifications not written out in full in this section, including the hazard classes and the hazard statements, the full text is listed in section 16.

3.2. Mixtures

Not applicable

SECTION 4: First-Aid Measures

4.1. Description of first aid measures

Immediately remove contaminated clothing.

If inhaled:
If difficulties occur after vapour/aerosol has been inhaled, remove to fresh air and seek medical attention.

On skin contact:
Wash thoroughly with soap and water

On contact with eyes:
Immediately wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Rinse mouth and then drink 200-300 ml of water. Do not induce vomiting unless told to by a poison control center or doctor.

4.2. Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far

4.3. Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

SECTION 5: Fire-Fighting Measures

5.1. Extinguishing media

Suitable extinguishing media:
dry powder, foam

5.2. Special hazards arising from the substance or mixture

Endangering substances: harmful vapours
Advice: Evolution of fumes/fog. The substances/groups of substances mentioned can be released in case of fire.

5.3. Advice for fire-fighters

Special protective equipment:
Wear a self-contained breathing apparatus.

Further information:
The degree of risk is governed by the burning substance and the fire conditions. Contaminated extinguishing water must be disposed of in accordance with official regulations.

SECTION 6: Accidental Release Measures

6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective clothing.

6.2. Environmental precautions

Contain contaminated water/firefighting water. Do not discharge into drains/surface waters/groundwater.

6.3. Methods and material for containment and cleaning up

For large amounts: Pump off product.
For residues: Pick up with suitable absorbent material. Dispose of absorbed material in accordance with regulations.

6.4. Reference to other sections

Information regarding exposure controls/personal protection and disposal considerations can be found in section 8 and 13.

SECTION 7: Handling and Storage

7.1. Precautions for safe handling

No special measures necessary provided product is used correctly.

Protection against fire and explosion:
Prevent electrostatic charge - sources of ignition should be kept well clear - fire extinguishers should be kept handy.

Temperature class: T2 (Autoignition temperature >300 °C).

7.2. Conditions for safe storage, including any incompatibilities

Further information on storage conditions: Keep container tightly closed and dry; store in a cool place.

Storage class according to TRGS 510 (originally VCI, Germany): (10) Combustible liquids

Storage stability:
If moisture enters isocyanate containers, CO2 forms and pressure builds up.

7.3. Specific end use(s)

See exposure scenario(s) in the attachment to this safety data sheet.

SECTION 8: Exposure Controls/Personal Protection

8.1. Control parameters

Components with occupational exposure limits

The surveillance of the workplace by exposure measurements may be necessary, in order to prove the efficiency of safety measures, for example ventilation or the need of respiratory protection. Since this requires a specific competency, only accredited laboratories should be contracted. Regarding suitable methods to assess inhalation exposure, the European Standards EN 482, 689 and 14042 are to be considered. In addition, the TRGS 402 has to be observed in Germany.

822-06-0: hexamethylene-di-isocyanate
Short Term Exposure Classification: (TRGS 900 (DE)), Vapor and aerosol
Category I: Substances for which the localized effect has an assigned exposure limit or for substances with a sensitizing effect in respiratory passages
Short Term Exposure Factor: (TRGS 900 (DE)), Vapor and aerosol
Ceiling limit value/factor: 1
Substance listed with exceeding factor and category of short time value.
OEL 0,035 mg/m3 ; 0,005 ppm (TRGS 900 (DE)), Vapor and aerosol
Ceiling limit value/factor: 2

PNEC

freshwater: 0,127 mg/l

marine water: 0,0127 mg/l

intermittent release: 1,27 mg/l

sediment (freshwater): 266700 mg/kg

marine water: 26670 mg/kg

soil: 53182 mg/kg

STP: 38,28 mg/l

DNEL

worker:
Short-term exposure - local effects, Inhalation: 1 mg/m3

worker:
Long-term exposure - local effects, Inhalation: 0,5 mg/m3

8.2. Exposure controls

Personal protective equipment

Respiratory protection:
Respiratory protection in case of vapour/aerosol release. Combination filter for gases/vapours of organic compounds and solid and liquid particles (f.e. EN 14387 Type A-P2)

Hand protection:
Chemical resistant protective gloves (EN ISO 374-1)
Suitable materials for short-term contact (recommended: At least protective index 2, corresponding > 30 minutes of permeation time according to EN ISO 374-1)
butyl rubber (butyl) - 0.7 mm coating thickness
Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures

Handle in accordance with good industrial hygiene and safety practice. Wearing of closed work clothing is required additionally to the stated personal protection equipment.

SECTION 9: Physical and Chemical Properties

9.1. Information on basic physical and chemical properties

State of matter:	liquid	
Form:	liquid	
Colour:	colourless to slightly yellow	
Odour:	faint odour	
Odour threshold:	Not determined due to potential health hazard by inhalation.	
Melting point:	-51,3 - -28,4 °C	(OECD Guideline 102)
Boiling range:	300 - 355 °C	
Flammability:	not flammable	(derived from flash point)
Lower explosion limit:	For liquids not relevant for classification and labelling., The lower explosion point may be 5 - 15 °C below the flash point.	
Upper explosion limit:	For liquids not relevant for classification and labelling.	
Flash point:	208 °C	(ISO 2719)
Auto-ignition temperature:	443 °C	(DIN EN 14522)
Thermal decomposition:	125 °C, 550 J/g (DSC (DIN 51007)) 405 °C, > 130 kJ/kg (DSC (DIN 51007)) 20 - 145 °C, 0 kJ/kg (Warm storage test) No exothermic decomposition within the mentioned temperature range.	
pH value:	not applicable	
Viscosity, dynamic:	2,5 - 4 Pa.s (23 °C, 1.000 1/s)	(DIN EN ISO 3219)
Solubility in water:	Reacts with water.	
Partitioning coefficient n-octanol/water (log Kow):	9,81	(calculated)
Vapour pressure:	0,00246 Pa (20 °C)	(Directive 92/69/EEC, A.4)
Relative density:	1,166 (20 °C)	
Density:	1,166 g/cm3 (20 °C)	(DIN 51757)
Relative vapour density (air):	not determined	

Particle characteristics

Particle size distribution: The substance / product is marketed or used in a non solid or granular form. -

9.2. Other information

Information with regard to physical hazard classes

Explosives

Explosion hazard: Based on the chemical structure there is no indication of explosive properties.

Impact sensitivity: Based on the chemical structure there is no shock-sensitivity.

Oxidizing properties

Fire promoting properties: Based on its structural properties the product is not classified as oxidizing.

Pyrophoric properties

Self-ignition temperature: Test type: Spontaneous self-ignition at room-temperature.

Based on its structural properties the product is not classified as self-igniting.

Substances and mixtures, which emit flammable gases in contact with water

Formation of flammable gases: Forms no flammable gases in the presence of water.

Other safety characteristics

Miscibility with water: Reacts with water.

Hygroscopy: hygroscopic

Adsorption/water - soil: log KOC: 7,3 - 7,8 (calculated)

Surface tension: Based on chemical structure, surface activity is not to be expected.

Evaporation rate: not determined

SECTION 10: Stability and Reactivity

10.1. Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

Formation of flammable gases:	Remarks:	Forms no flammable gases in the presence of water.

10.2. Chemical stability

The product is stable if stored and handled as prescribed/indicated.

10.3. Possibility of hazardous reactions

Reacts with alcohols. Reacts with amines. Reacts with substances which contain active hydrogen. Reacts with water, with formation of carbon dioxide. The formation of gaseous decomposition products builds up pressure in tightly closed containers.

10.4. Conditions to avoid

Avoid moisture. See SDS section 7 - Handling and storage.

10.5. Incompatible materials

Substances to avoid:
water, alcohols, amines

10.6. Hazardous decomposition products

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

SECTION 11: Toxicological Information

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Acute toxicity

Assessment of acute toxicity:
Virtually nontoxic after a single ingestion. Virtually nontoxic after a single skin contact. Of moderate toxicity after short-term inhalation.

Experimental/calculated data:
LD50 rat (oral): > 2.500 mg/kg (OECD Guideline 423)
No mortality was observed.

LC50 rat (by inhalation): 0,467 mg/l 4 h (OECD Guideline 403)
The test result applies only to the substance transferred into respirable aerosol (particles < 20 µm).
An aerosol was tested.

LD50 rat (dermal): > 2.000 mg/kg (OECD Guideline 402)

Irritation

Assessment of irritating effects:
May cause slight irritation to the skin. May cause slight irritation to the eyes.

Experimental/calculated data:
Skin corrosion/irritation
rabbit: non-irritant (OECD Guideline 404)

Serious eye damage/irritation
rabbit: non-irritant (OECD Guideline 405)

Respiratory/Skin sensitization

Assessment of sensitization:
Caused skin sensitization in animal studies.

Experimental/calculated data:
Guinea pig maximization test guinea pig: skin sensitizing
sensitizing effect in animal tests

Germ cell mutagenicity

Assessment of mutagenicity:
The substance was not mutagenic in bacteria. The substance was not mutagenic in mammalian cell culture.

Carcinogenicity

Assessment of carcinogenicity:
Study scientifically not justified.

Reproductive toxicity

Assessment of reproduction toxicity:
Study scientifically not justified.

Developmental toxicity

Assessment of teratogenicity:
Study scientifically not justified.

Specific target organ toxicity (single exposure)

Assessment of STOT single:
Causes temporary irritation of the respiratory tract.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
After repeated exposure the prominent effect is local irritation.

Aspiration hazard

No aspiration hazard expected.

Interactive effects

No data available.

11.2. Information on other hazards

Endocrine disrupting properties

The substance is not identified to have endocrine disrupting properties according to Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 nor is included in the Candidate List of substances of very high concern according to EU REACh Article 59 for having endocrine disrupting properties.

SECTION 12: Ecological Information

12.1. Toxicity

Toxicity to fish:
LC0 (96 h) >= 100 mg/l, Brachydanio rerio (OECD 203; ISO 7346; 92/69/EEC, C.1, static)
The product may hydrolyse. The test result maybe partially due to degradation products. The product has low solubility in the test medium. An eluate has been tested. Nominal concentration.

Aquatic invertebrates:
EL50 (48 h) 127 mg/l, Daphnia magna (Directive 92/69/EEC, C.2, static)

Nominal concentration. The product may hydrolyse. The test result maybe partially due to degradation products.

Aquatic plants:
EC50 (72 h) > 1.000 mg/l (growth rate), Scenedesmus subspicatus (OECD Guideline 201, static)
Nominal concentration. The product may hydrolyse. The test result maybe partially due to degradation products.

Microorganisms/Effect on activated sludge:
EC20 (3 h) 880 mg/l, (OECD Guideline 209, static)
Nominal concentration. The product may hydrolyse. The test result maybe partially due to degradation products.

Chronic toxicity to fish:
Study not necessary due to exposure considerations.

Chronic toxicity to aquatic invertebrates:
Study not necessary due to exposure considerations.

Assessment of terrestrial toxicity:
No data available concerning terrestrial toxicity.
Study not necessary due to exposure considerations.

12.2. Persistence and degradability

Assessment biodegradation and elimination (H2O):
Poorly biodegradable. The product is unstable in water. The elimination data also refer to products of hydrolysis.

Elimination information:
1 % BOD of the ThOD (28 d) (OECD 301D; EEC 92/69, C.4-E) (aerobic, activated sludge, domestic, non-adapted)

Assessment of stability in water:
In contact with water the substance will hydrolyse rapidly.

Information on Stability in Water (Hydrolysis):
$t_{1/2}$ < 1 h, (OECD Guideline 111, pH 4)

$t_{1/2}$ < 1 h, (OECD Guideline 111, pH 7)

$t_{1/2}$ < 1 h, (OECD Guideline 111, pH 9)

12.3. Bioaccumulative potential

Assessment bioaccumulation potential:
Significant accumulation in organisms is not to be expected.

The product has not been tested. The statement has been derived from the structure of the product.

Bioaccumulation potential:

Bioconcentration factor(BCF): 367,7, Fish (calculated)
Analogous: Assessment derived from products with similar chemical character.

12.4. Mobility in soil

Assessment transport between environmental compartments:
Volatility: The substance will not evaporate into the atmosphere from the water surface.
Adsorption in soil: No data available.

12.5. Results of PBT and vPvB assessment

According to Annex XIII of Regulation (EC) No.1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH): The product does not fulfill the criteria for PBT (Persistent/bioaccumulative/toxic) and vPvB (very persistent/very bioaccumulative). Self classification

12.6. Endocrine disrupting properties

The substance is not identified to have endocrine disrupting properties according to Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 nor is included in the Candidate List of substances of very high concern according to EU REACh Article 59 for having endocrine disrupting properties.

12.7. Other adverse effects

The product does not contain substances that are listed in Annex I of Regulation (EC) 2037/2000 on substances that deplete the ozone layer.

12.8. Additional information

Other ecotoxicological advice:
Do not release untreated into natural waters. The local regulations on waste-water treatment must be followed.

SECTION 13: Disposal Considerations

13.1. Waste treatment methods

Must be disposed of or incinerated in accordance with local regulations.

Contaminated packaging:
Uncontaminated packaging can be re-used.
Packs that cannot be cleaned should be disposed of in the same manner as the contents.

SECTION 14: Transport Information

Land transport

ADR

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

RID

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Inland waterway transport
ADN

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user:	None known

Transport in inland waterway vessel
Not evaluated

Sea transport

IMDG

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Air transport

IATA/ICAO

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

14.1. UN number or ID number
See corresponding entries for “UN number or ID number” for the respective regulations in the tables above.

14.2. UN proper shipping name
See corresponding entries for “UN proper shipping name” for the respective regulations in the tables above.

14.3. Transport hazard class(es)
See corresponding entries for “Transport hazard class(es)” for the respective regulations in the tables above.

14.4. Packing group
See corresponding entries for “Packing group” for the respective regulations in the tables above.

14.5. Environmental hazards
See corresponding entries for “Environmental hazards” for the respective regulations in the tables above.

14.6. Special precautions for user

See corresponding entries for “Special precautions for user” for the respective regulations in the tables above.

14.7. Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

SECTION 15: Regulatory Information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

Prohibitions, Restrictions and Authorizations

Annex XVII of Regulation (EC) No 1907/2006: Number on List: 3, 74, 75

Hazardous Incident Ordinance (Germany):
Listed in above regulation: no

Directive 2012/18/EU - Control of Major Accident Hazards involving dangerous substances (EU):
Listed in above regulation: no

Water hazard class (§6 AwSV para.4 (Legal binding announcement of the substance in the Federal Gazette)): (1) Weakly water polluting. ID-No.: 6721

TRGS 430 'Isocyanates - risk assessment and safety precautions' (Germany)
The specifications of the Technical Rule for Hazardous Substances (TRGS) 401 must be observed (TRGS 401: Risks resulting from skin contact - identification, assessment, measures).
Law on the Protection of Working Youth
The Maternity Protection Act needs to be considered.

15.2. Chemical Safety Assessment

Chemical Safety Assessment performed

SECTION 16: Other Information

This product is of industrial quality and unless otherwise specified or agreed intended exclusively for industrial use. This includes the mentioned and recommended usage. Any other intended applications should be discussed with the manufacturer. In particular this concerns the application for products that are the object of special standards and regulations.

The product is used mainly as a hardener in coating materials or adhesives. The handling of coating materials or adhesives containing reactive polyisocyanates and residual monomeric HDI requires appropriate protective measures referred to in this safety data sheet. These products may therefore be used only in industrial or trade applications. They are not suitable for use in homeworker (DIY) applications.

Acute Tox.	Acute toxicity
Skin Sens.	Skin sensitization
STOT SE	Specific target organ toxicity — single exposure
Skin Corr./Irrit.	Skin corrosion/irritation
Eye Dam./Irrit.	Serious eye damage/eye irritation
Resp. Sens.	Respiratory sensitization
H332	Harmful if inhaled.
H317	May cause an allergic skin reaction.
H335	May cause respiratory irritation.
H319	Causes serious eye irritation.
H315	Causes skin irritation.
H330	Fatal if inhaled.
H302	Harmful if swallowed.
H334	May cause allergy or asthma symptoms or breathing difficulties if inhaled.

Abbreviations
ADR = The European Agreement concerning the International Carriage of Dangerous Goods by Road. ADN = The European Agreement concerning the International Carriage of Dangerous Goods by Inland waterways. ATE = Acute Toxicity Estimates. CAO = Cargo Aircraft Only. CAS = Chemical Abstract Service. CLP = Classification, Labelling and Packaging of substances and mixtures. DIN = German national organization for standardization. DNEL = Derived No Effect Level. EC50 = Effective concentration median for 50% of the population. EC = European Community. EN = European Standards. IARC = International Agency for Research on Cancer. IATA = International Air Transport Association. IBC-Code = Intermediate Bulk Container code. IMDG = International Maritime Dangerous Goods Code. ISO = International Organization for Standardization. STEL = Short-Term Exposure Limit. LC50 = Lethal concentration median for 50% of the population. LD50 = Lethal dose median for 50% of the population. TLV = Threshold Limit Value. MARPOL = The International Convention for the Prevention of Pollution from Ships. NEN = Dutch Norm. NOEC = No Observed Effect Concentration. OEL = Occupational Exposure Limit. OECD = Organization for Economic Cooperation and Development. PBT = Persistent, Bioaccumulative and Toxic. PNEC = Predicted No Effect Level. PPM = Parts per million. RID = The European Agreement concerning the International Carriage of Dangerous Goods by Rail. TWA = Time Weight Average. UN-number = UN number at transport. vPvB = very Persistent and very Bioaccumulative.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Annex: Exposure Scenarios

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3. Professional applications
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1. Short title of exposure scenario

Formulation
SU3, SU10; ERC2; PROC1, PROC2, PROC3, PROC4, PROC5, PROC8a, PROC8b, PROC9, PROC15

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed.

Contributing exposure scenario	
Use descriptors covered	PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions. PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition PROC4: Chemical production where opportunity for exposure arises PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). PROC15: Use a laboratory reagent. Use domain: industrial

Operational conditions	
Concentration of the substance	HDI oligomers, isocyanurate Content: >= 0 % - <= 100 %
Physical state	Liquid, very low fugacity
Duration and Frequency of activity	Exposure duration: 480 min 5 days per week
Indoor/Outdoor	Indoor
Risk Management Measures	
Regular inspection and maintenance of equipment and machines. Avoid frequent and direct contact with substance. Clean equipment and the work area every day.	
Engineering controls have to be used to reduce exposures.	
Wear suitable face shield Wear suitable coveralls to prevent exposure to the skin. Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Local exhaust ventilation	Effectiveness: 90,0 %
Wear chemically resistant gloves in combination with ‘basic’ employee training.	Effectiveness: 90,0 %
Change gloves, if duration of activity exceeds break through time	
Ensure material transfers are under containment or extract ventilation	
Relevant for long-term processes	
Relevant for short-term processes, Wear an air-fed mask., Alternatively:, Wear a mask with an activated carbon filter combined with a particulate filter.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - dermal
Assessment method	ECETOC TRA v2.0 Worker
	Worker - inhalation, long-term - local
Exposure estimate	0,21 mg/m³
Risk Characterization Ratio (RCR)	0,42
	The short-term exposure is covered by the assessment of long-term exposure.
Additional good practice advice	
Ensure good work practices are implemented. Avoid handling of the substance in case of known skin complaints, hypersensitivity reactions, chronic respiratory disease, asthmatic attacks or	

bronchial attacks. Workers should receive a pre-placement examination and subsequent periodic medical examinations, including a pulmonary function test.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC5: Mixing or blending in batch processes PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities Use domain: industrial
Operational conditions	
Concentration of the substance	HDI oligomers, isocyanurate Content: >= 0 % - <= 100 %
Physical state	Liquid, very low fugacity
Duration and Frequency of activity	Exposure duration: 480 min 5 days per week
Indoor/Outdoor	Indoor
Risk Management Measures	
Regular inspection and maintenance of equipment and machines. Avoid frequent and direct contact with substance. Clean equipment and the work area every day.	
Engineering controls have to be used to reduce exposures.	
Use suitable eye protection. Wear suitable coveralls to prevent exposure to the skin. Wear suitable face shield	
Risk Management Measures are based on qualitative risk characterisation.	
Local exhaust ventilation	Effectiveness: 90,0 %
Wear chemically resistant gloves in combination with ‘basic’ employee training.	Effectiveness: 90,0 %
Change gloves, if duration of activity exceeds break through time	
Ensure material transfers are under containment or extract ventilation	
Relevant for long-term processes	
Relevant for short-term processes, Wear an air-fed mask., Alternatively:, Wear a mask with an activated carbon filter combined with a particulate filter.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment

	Worker - dermal
Assessment method	ECETOC TRA v2.0 Worker
	Worker - inhalation, long-term - local
Exposure estimate	0,21 mg/m³
Risk Characterization Ratio (RCR)	0,42
	The short-term exposure is covered by the assessment of long-term exposure.
Additional good practice advice	
Ensure good work practices are implemented. Avoid handling of the substance in case of known skin complaints, hypersensitivity reactions, chronic respiratory disease, asthmatic attacks or bronchial attacks. Workers should receive a pre-placement examination and subsequent periodic medical examinations, including a pulmonary function test.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

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2. Short title of exposure scenario

Industrial applications
SU3, SU12, SU13, SU19; ERC5, ERC6a, ERC6c, ERC6d; PROC1, PROC2, PROC3, PROC4, PROC5, PROC7, PROC8a, PROC8b, PROC9, PROC10, PROC13, PROC14, PROC15

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed.

Contributing exposure scenario	
Use descriptors covered	PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions. PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Use domain: industrial
Operational conditions	
Concentration of the substance	HDI oligomers, isocyanurate Content: >= 0 % - <= 50 %
Physical state	Liquid, very low fugacity
Duration and Frequency of activity	Exposure duration: 480 min 5 days per week
Duration and Frequency of activity	Exposure duration: 240 min 5 days per week

	Relevant for PROC 8a
Indoor/Outdoor	Indoor
Risk Management Measures	
Regular inspection and maintenance of equipment and machines. Avoid frequent and direct contact with substance. Clean equipment and the work area every day.	
Engineering controls have to be used to reduce exposures.	
Wear suitable face shield Wear suitable coveralls to prevent exposure to the skin. Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Local exhaust ventilation	Effectiveness: 90,0 %
Wear chemically resistant gloves in combination with ‘basic’ employee training.	Effectiveness: 90,0 %
Change gloves, if duration of activity exceeds break through time	
Ensure material transfers are under containment or extract ventilation Ensure samples are obtained under containment or extract ventilation.	
Relevant for long-term processes	
Relevant for short-term processes, Wear an air-fed mask., Alternatively:, Wear a mask with an activated carbon filter combined with a particulate filter.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - dermal
Assessment method	ECETOC TRA v2.0 Worker
	Worker - inhalation, long-term - local
Exposure estimate	0,21 mg/m³
Risk Characterization Ratio (RCR)	0,42
	The short-term exposure is covered by the assessment of long-term exposure.
Additional good practice advice	
Ensure good work practices are implemented. Avoid handling of the substance in case of known skin complaints, hypersensitivity reactions, chronic respiratory disease, asthmatic attacks or bronchial attacks. Workers should receive a pre-placement examination and subsequent periodic medical examinations, including a pulmonary function test.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC5: Mixing or blending in batch processes PROC10: Roller application or brushing Use domain: industrial
Operational conditions	
Concentration of the substance	HDI oligomers, isocyanurate Content: >= 0 % - <= 50 %
Physical state	Liquid, very low fugacity
Duration and Frequency of activity	Exposure duration: 480 min 5 days per week Relevant for PROC 10
Duration and Frequency of activity	Exposure duration: 240 min 5 days per week Relevant for PROC 5
Indoor/Outdoor	Indoor
Risk Management Measures	
Regular inspection and maintenance of equipment and machines. Avoid frequent and direct contact with substance. Clean equipment and the work area every day.	
Engineering controls have to be used to reduce exposures. Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings.	
Wear suitable face shield Wear an impervious suit. Use suitable eye protection.	
Wear an air-fed mask., Risk Management Measures are based on qualitative risk characterisation.	
Local exhaust ventilation	Effectiveness: 90,0 %
Wear chemically resistant gloves in combination with ‘basic’ employee training.	Effectiveness: 90,0 %
Change gloves, if duration of activity exceeds break through time	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - dermal
Assessment method	ECETOC TRA v2.0 Worker
	Worker - inhalation, long-term - local
Exposure estimate	0,21 mg/m³
Risk Characterization Ratio (RCR)	0,42
	The short-term exposure is covered by the assessment of long-term exposure.

Additional good practice advice
Ensure good work practices are implemented. Avoid handling of the substance in case of known skin complaints, hypersensitivity reactions, chronic respiratory disease, asthmatic attacks or bronchial attacks. Workers should receive a pre-placement examination and subsequent periodic medical examinations, including a pulmonary function test.
Guidance to Downstream Users
For scaling see: http://www.ecetoc.org/tra

Contributing exposure scenario	
Use descriptors covered	PROC7: Industrial spraying Use domain: industrial
Operational conditions	
Concentration of the substance	HDI oligomers, isocyanurate Content: >= 0 % - <= 50 %
Physical state	Liquid, very low fugacity
Duration and Frequency of activity	Exposure duration: 240 min 5 days per week
Indoor/Outdoor	Indoor
Risk Management Measures	
Regular inspection and maintenance of equipment and machines. Avoid frequent and direct contact with substance. Clean equipment and the work area every day.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Engineering controls have to be used to reduce exposures.	
Wear suitable face shield Wear an impervious suit. Use suitable eye protection.	
Wear an air-fed mask., Risk Management Measures are based on qualitative risk characterisation.	
Local exhaust ventilation	Effectiveness: 90,0 %
Wear chemically resistant gloves in combination with ‘basic’ employee training.	Effectiveness: 90,0 %
Change gloves, if duration of activity exceeds break through time	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - dermal
Assessment method	ECETOC TRA v2.0 Worker

	Worker - inhalation, long-term - local
Exposure estimate	0,21 mg/m³
Risk Characterization Ratio (RCR)	0,42
	The short-term exposure is covered by the assessment of long-term exposure.
Additional good practice advice	
Ensure good work practices are implemented. Avoid handling of the substance in case of known skin complaints, hypersensitivity reactions, chronic respiratory disease, asthmatic attacks or bronchial attacks. Workers should receive a pre-placement examination and subsequent periodic medical examinations, including a pulmonary function test.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition PROC4: Chemical production where opportunity for exposure arises PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). PROC13: Treatment of articles by dipping and pouring. PROC14: Tabletting, compression, extrusion, pelletisation, granulation PROC15: Use a laboratory reagent. Use domain: industrial and professional
Operational conditions	
Concentration of the substance	HDI oligomers, isocyanurate Content: >= 0 % - <= 50 %
Physical state	Liquid, very low fugacity
Duration and Frequency of activity	Exposure duration: 480 min 5 days per week
Duration and Frequency of activity	Exposure duration: 240 min 5 days per week Relevant for PROC 8a
Indoor/Outdoor	Indoor
Risk Management Measures	
Regular inspection and maintenance of equipment and machines. Avoid frequent and direct contact with substance. Clean equipment and the work area every day.	
Engineering controls have to be used to reduce exposures.	
Use suitable eye protection. Wear suitable face shield Wear suitable	

coveralls to prevent exposure to the skin.	
Risk Management Measures are based on qualitative risk characterisation.	
Local exhaust ventilation	Effectiveness: 90,0 %
Wear chemically resistant gloves in combination with ‘basic’ employee training.	Effectiveness: 90,0 %
Change gloves, if duration of activity exceeds break through time	
Ensure material transfers are under containment or extract ventilation	
Relevant for long-term processes	
Relevant for short-term processes, Wear an air-fed mask., Alternatively:, Wear a mask with an activated carbon filter combined with a particulate filter.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - dermal
Assessment method	ECETOC TRA v2.0 Worker
	Worker - inhalation, long-term - local
Exposure estimate	0,21 mg/m³
Risk Characterization Ratio (RCR)	0,42
	The short-term exposure is covered by the assessment of long-term exposure.
Additional good practice advice	
Ensure good work practices are implemented. Avoid handling of the substance in case of known skin complaints, hypersensitivity reactions, chronic respiratory disease, asthmatic attacks or bronchial attacks. Workers should receive a pre-placement examination and subsequent periodic medical examinations, including a pulmonary function test.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

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3. Short title of exposure scenario

Professional applications
SU22, SU10, SU12, SU13, SU19; ERC2, ERC8c, ERC8f; PROC3, PROC4, PROC5, PROC8a, PROC8b, PROC9, PROC10, PROC11, PROC13, PROC14, PROC15

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	As no environmental hazard was identified no environmental-related exposure assessment and risk

	characterization was performed.

Contributing exposure scenario	
Use descriptors covered	PROC5: Mixing or blending in batch processes PROC10: Roller application or brushing Use domain: industrial
Operational conditions	
Concentration of the substance	HDI oligomers, isocyanurate Content: >= 0 % - <= 50 %
Physical state	Liquid, very low fugacity
Duration and Frequency of activity	Exposure duration: 480 min 5 days per week Relevant for PROC 10
Duration and Frequency of activity	Exposure duration: 240 min 5 days per week Relevant for PROC 5
Indoor/Outdoor	Indoor
Risk Management Measures	
Regular inspection and maintenance of equipment and machines. Avoid frequent and direct contact with substance. Clean equipment and the work area every day.	
Engineering controls have to be used to reduce exposures. Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings.	
Wear suitable face shield Wear an impervious suit. Use suitable eye protection.	
Wear an air-fed mask., Risk Management Measures are based on qualitative risk characterisation.	
Local exhaust ventilation	Effectiveness: 90,0 %
Wear chemically resistant gloves in combination with ‘basic’ employee training.	Effectiveness: 90,0 %
Change gloves, if duration of activity exceeds break through time	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - dermal
Assessment method	ECETOC TRA v2.0 Worker
	Worker - inhalation, long-term - local
Exposure estimate	0,21 mg/m³
Risk Characterization Ratio (RCR)	0,42

	The short-term exposure is covered by the assessment of long-term exposure.
Additional good practice advice	
Ensure good work practices are implemented. Avoid handling of the substance in case of known skin complaints, hypersensitivity reactions, chronic respiratory disease, asthmatic attacks or bronchial attacks. Workers should receive a pre-placement examination and subsequent periodic medical examinations, including a pulmonary function test.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC11: Non industrial spraying Use domain: industrial
Operational conditions	
Concentration of the substance	HDI oligomers, isocyanurate Content: >= 0 % - <= 50 %
Physical state	Liquid, very low fugacity
Duration and Frequency of activity	Exposure duration: 480 min 5 days per week
Indoor/Outdoor	Indoor
Risk Management Measures	
Regular inspection and maintenance of equipment and machines. Avoid frequent and direct contact with substance. Clean equipment and the work area every day.	
Engineering controls have to be used to reduce exposures. Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings.	
Wear suitable face shield Wear an impervious suit. Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation., Wear an air-fed mask.	
Local exhaust ventilation	Effectiveness: 90 %
Wear chemically resistant gloves in combination with ‘basic’ employee training.	Effectiveness: 90,0 %
Change gloves, if duration of activity exceeds break through time	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment

	Worker - dermal
Assessment method	ECETOC TRA v2.0 Worker
	Worker - inhalation, long-term - local
Exposure estimate	0,21 mg/m³
Risk Characterization Ratio (RCR)	0,42
	The short-term exposure is covered by the assessment of long-term exposure.
Additional good practice advice	
Ensure good work practices are implemented. Avoid handling of the substance in case of known skin complaints, hypersensitivity reactions, chronic respiratory disease, asthmatic attacks or bronchial attacks. Workers should receive a pre-placement examination and subsequent periodic medical examinations, including a pulmonary function test.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition PROC4: Chemical production where opportunity for exposure arises PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). PROC13: Treatment of articles by dipping and pouring. PROC14: Tabletting, compression, extrusion, pelletisation, granulation PROC15: Use a laboratory reagent. Use domain: industrial and professional
Operational conditions	
Concentration of the substance	HDI oligomers, isocyanurate Content: >= 0 % - <= 50 %
Physical state	Liquid, very low fugacity
Duration and Frequency of activity	Exposure duration: 480 min 5 days per week
Duration and Frequency of activity	Exposure duration: 240 min 5 days per week Relevant for PROC 8a
Indoor/Outdoor	Indoor
Risk Management Measures	
Regular inspection and maintenance of equipment and machines. Avoid frequent and direct contact with substance. Clean equipment and the work area every day.	
Engineering controls have to be used to reduce exposures.	

Use suitable eye protection. Wear suitable face shield Wear suitable coveralls to prevent exposure to the skin.	
Risk Management Measures are based on qualitative risk characterisation.	
Local exhaust ventilation	Effectiveness: 90,0 %
Wear chemically resistant gloves in combination with ‘basic’ employee training.	Effectiveness: 90,0 %
Change gloves, if duration of activity exceeds break through time	
Ensure material transfers are under containment or extract ventilation	
Relevant for long-term processes	
Relevant for short-term processes, Wear an air-fed mask., Alternatively:, Wear a mask with an activated carbon filter combined with a particulate filter.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - dermal
Assessment method	ECETOC TRA v2.0 Worker
	Worker - inhalation, long-term - local
Exposure estimate	0,21 mg/m³
Risk Characterization Ratio (RCR)	0,42
	The short-term exposure is covered by the assessment of long-term exposure.
Additional good practice advice	
Ensure good work practices are implemented. Avoid handling of the substance in case of known skin complaints, hypersensitivity reactions, chronic respiratory disease, asthmatic attacks or bronchial attacks. Workers should receive a pre-placement examination and subsequent periodic medical examinations, including a pulmonary function test.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

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